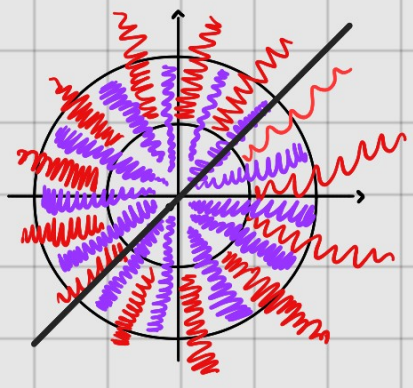
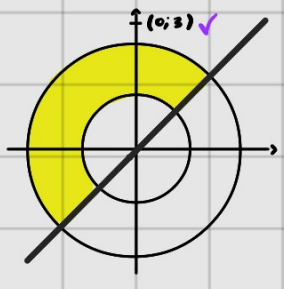
$$(\rho\cos\theta)^2+(\rho\operatorname{sen}\theta)^2=1$$
$$\rho^2\cos^2\theta+\rho^2\operatorname{sen}^2\theta=1$$

PROBLEMA 3

CALCOLARE IL SEGUENTE INTEGRALE DOPPIO: $\iint_C 4xy(x^2+y^2)\,dx\,dy$ CON $C=\{(x,y)\in\mathbb{R}^2 \mid 1\le x^2+y^2\le 4,\ y\ge x\}$



- $x^2+y^2=1$ DEVO PRENDERE LA PARTE ESTERNA ALLA CIRCONFERENZA PIÚ PICCOLA CIRCONFERENZA DI CENTRO L'ORIGINE E RAGGIO 1
- $x^2+y^2=4$ DEVO PRENDERE LA PARTE INTERNA ALLA CIRCONFERENZA DI RAGGIO 2 CIRCONFERENZA DI CENTRO L'ORIGINE E RAGGIO 2
- $y=x$ RETTA (BISETTRICE DEL PRIMO E TERZO QUADRANTE)



COME FACCIO A CAPIRE SE PRENDERE LA PARTE SUP O INF ALLA BISETTRICE?
- PRENDIAMO UN PUNTO A CASO (0;3) FA PARTE DELLA PARTE DEL SEMIPIANO GIUSTA?
- PONGO $y\ge x \longrightarrow 3\ge 0$ SI ✓

PARAMETRIZZIAMO IL DOMINIO CON COORDINATE POLARI:
PERCHÈ IN COORDINATE POLARI?
PERCHÈ OGNI VOLTA CHE AVETE UN DOMINIO CHE HA UNA SIMMETRIA SFERICA, CIOÈ PARTE DI CIRCONFERENZE, SETTORI DI CIRCONFERENZE

- $x=\rho\cos\theta$ ρ SAREBBE IL NOSTRO RAGGIO QUINDI $[1,2]$
- $y=\rho\operatorname{sen}\theta$ θ IN $[\frac{\pi}{4},\frac{5}{4}\pi]$

$$\iint_C 4(\rho\cos\theta)(\rho\operatorname{sen}\theta)\rho^2\cdot\rho\,d\rho\,d\theta$$ DEVONO ESSE MOLTIPLICATI PER LA MATRICE JACOBIANA

$$\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} d\theta\int_1^2 4\rho^5\operatorname{sen}\theta\cos\theta\,d\rho=\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} d\theta\left[\frac{2}{6}\frac{\rho^6}{3}\operatorname{sen}\theta\cos\theta\right]_1^2=\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}}\operatorname{sen}\theta\cos\theta\cdot\frac{2}{3}(64-1)\,d\theta=\frac{126}{3}\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}}\operatorname{sen}\theta\cos\theta\,d\theta$$

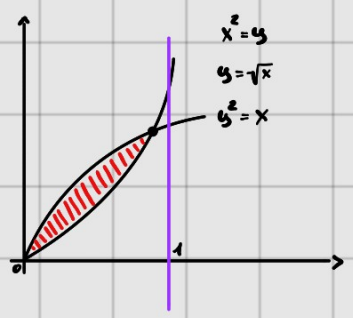
$$=\frac{126}{3}\left[\frac{\operatorname{sen}^2\theta}{2}\right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}=\frac{126}{3}\left(\frac{1}{4}-\frac{1}{4}\right)=0$$

INTEGRALI DOPPI

DOMINIO NORMALE RISPETTO ASSE x	DOMINIO NORMALE RISPETTO ASSE y
$a\le x\le b$ $\quad\gamma(x)\le y\le\delta(x)$	$\alpha(y)\le x\le\beta(y)$ $\quad c\le y\le d$
$\iint_\Omega f(x,y)\,dx\,dy=\int_a^b dx\int_{\gamma(x)}^{\delta(x)} f(x,y)\,dy$	$\iint_\Omega g(x,y)\,dx\,dy=\int_c^d dy\int_{\alpha(y)}^{\beta(y)} g(x,y)\,dx$

ESERCIZIO

$$\iint_E 4x^3y\,dx\,dy \qquad E=\{0\le x\le 1,\ x^2\le y\le\sqrt{x}\}$$



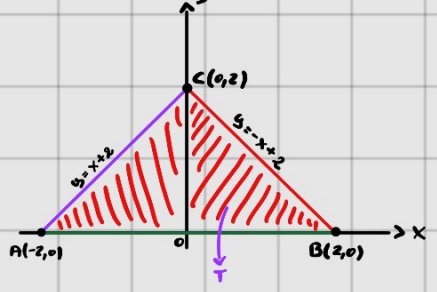
E È NORMALE RISPETTO ASSE x

$$\int_0^1 dx\int_{x^2}^{\sqrt{x}} 4x^3y\,dy=\int_0^1\left[\frac{2}{4}x^3\cdot\frac{y^2}{2}\right]_{x^2}^{\sqrt{x}}dx=\int_0^1(2x^3\cdot x-2x^3\cdot x^4)\,dx=$$

$$=\int_0^1 2x^4-2x^7\,dx=\left[2\frac{x^5}{5}-2\frac{x^8}{8}\right]_0^1=\left(\frac{2}{5}-\frac{1}{4}\right)-(0-0)=\frac{8-5}{20}=\frac{3}{20}$$

ESERCIZIO

$\iint_T x\cdot y^2 + e^x \, dx\,dy$ $\quad T = \overset{\Delta}{ABC}$, $A=(-2,0)$, $B=(2,0)$, $C=(0,2)$



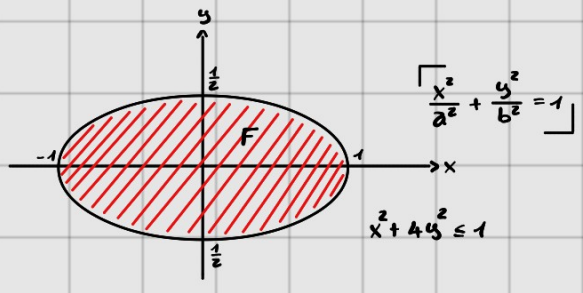
$T = \{0 \le y \le 2,\ y-2 \le -y+2\}$

$$\int_0^2 dy \int_{y-2}^{-y+2} x y^2 + e^x \, dx = \int_0^2 dy \left[\frac{x^2}{2} y^2 + e^x\right]_{y-2}^{-y+2} =$$

$$\int_0^2 \left(\frac{(-y+2)}{2} y^2 + e^{-y+2} - \left(\frac{(y-2)^2}{2} y^2 + e^{y-2} \right) \right) dy = \int_0^2 \left(\frac{y^2-2y+4}{2} y^2 + e^{-y+2} - \frac{y^2-2y+4}{2} y^2 - e^{y-2} \right) dy = \left(-e^{-2+2} - e^{2-2}\right) - \left(-e^{-0+2} - e^{0-2}\right) = -1-1+e^2+e^{-2} = -2+e^2+e^{-2}$$

ESERCIZIO

$\iint_F 1 \, dx\,dy = \text{AREA}(F)$ $\qquad F = \{x^2 + 4y^2 \le 1\}$



$\left[\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1\right]$

$x^2 + 4y^2 \le 1$

COORDINATE POLARI (GENERALIZZATE)

$$\begin{cases} x = \rho\cos\theta & \theta \in [0, 2\pi] \\ y = \frac{1}{2}\rho\sin\theta & \rho \in [0,1] \end{cases}$$

COORDINATE POLARI CLASSICHE PARTO DAL CERCHIO $x^2 + y^2 \le r^2$

$$\begin{cases} x = \rho\cos\theta & \theta \in [0, 2\pi] \\ y = \rho\sin\theta & \rho \in [0, r] \end{cases}$$

COME CAMBIA L'INTEGRALE ?

SOSTITUISCO x, y CON LE NUOVE VARIABILI, POI CALCOLO $\text{Jac} = \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -\rho\sin\theta \\ \frac{1}{2}\sin\theta & \frac{1}{2}\rho\cos\theta \end{vmatrix}$

INFINE $dx\,dy$ DIVENTA $|\det(\text{Jac})| \cdot d\rho\, d\theta$

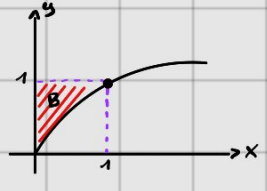
$$\det(\text{Jac}) = \cos\theta \cdot \frac{1}{2}\rho\cos\theta - \left(-\rho\sin\theta \cdot \frac{1}{2}\sin\theta\right) = \frac{1}{2}\rho\cos^2\theta + \frac{1}{2}\rho\sin\theta = \frac{1}{2}\rho\left(\cancel{\cos^2\theta + \sin^2\theta}^{\,1}\right) = \frac{1}{2}\rho \qquad F = \{(\rho,\theta) \in [0,1]\times[0,2\pi]\}$$

$$\Longrightarrow \iint_F 1\,dx\,dy = \iint_F \frac{1}{2}\rho\, d\rho\, d\theta = \int_0^{2\pi} d\theta \int_0^1 \frac{1}{2}\rho\, d\rho = \int_0^{2\pi} d\theta \left[\frac{1}{2}\cdot\frac{\rho^2}{2}\right]_0^1 = \int_0^{2\pi} \left(\frac{1}{4} - 0\right) d\theta = \frac{1}{4}\int_0^{2\pi} d\theta = \frac{1}{4}\cdot 2\pi = \frac{2\pi}{4} = \frac{\pi}{2}$$

ESERCIZIO

$\iint_B \frac{y}{(1+x)(1+y^2)}\,dx\,dy$

DOVE LA CURVA "INFERIORE" È $y = \sqrt{x}$



NORMALE RISPETTO ALL'ASSE X $\quad 0 \le x \le 1 \quad \sqrt{x} \le y \le 1$

$$\int_0^1 dx \int_{\sqrt{x}}^1 \frac{y}{(1+x)(1+y^2)}\,dy = \int_0^1 \frac{1}{1+x}\,dx \int \frac{y}{1+y^2}\,dy = \qquad \begin{matrix} u = 1+y^2 \\ du = 2y\,dy \end{matrix}$$

$$\int_0^1 \frac{1}{1+x}\,dx \cdot \frac{1}{2}\left[\ln(1+1^2) - \ln\left(1+(\cancel{\sqrt{x}}^{\,x})^2\right)\right]$$

$$\int_0^1 \frac{1}{1+x} \cdot \frac{1}{2}\left[\ln(2) - \ln(1+x)\right]dx = \frac{1}{2}\int_0^1 \frac{\ln(2)}{1+x}\,dx - \frac{1}{2}\int_0^1 \frac{\ln(1+x)}{1+x}\,dx$$

$$\int_0^1 \frac{\ln(2)}{1+x}\,dx = \ln(2)\int_0^1 \frac{1}{1+x}\,dx = \ln(2)\ln(1+x)\Big|_0^1 = \ln(2)(\ln 2)$$

$u = 1+x$ $\qquad \int_0^1 \frac{\ln(1+x)}{1+x}\,dx = \frac{1}{2}\ln^2(2)$

$$\int_0^1 dx \int_{\sqrt{x}}^1 \frac{y}{(1+x)(1+y^2)}\,dy = \frac{1}{2}\left[\ln^2(2) - \frac{1}{2}\ln^2(2)\right] = \frac{1}{4}\ln^2(2)$$

NORMALE RISPETTO ALL'ASSE Y $\quad 0 \le x \le y^2 \quad 0 \le y \le 1$

$$\int_0^1 dy \int_0^{y^2} \frac{y}{(1+x)(1+y^2)}\,dx = \int_0^1 \frac{y}{1+y^2}\,dy \int_0^{y^2} \frac{1}{1+x}\,dx =$$

$$\int_0^{y^2} \frac{1}{1+x}\,dx = \ln(1+x)\Big|_0^{y^2} = \ln(1+y^2) - \ln(1) = \ln(1+y^2)$$

$$\int_0^1 \frac{y}{1+y^2} \cdot \ln(1+y^2)\,dy = \frac{1}{2}\int_1^2 \frac{\ln(u)}{u}\,du \longrightarrow \int \frac{\ln(u)}{u}\,du = \frac{\ln^2(u)}{2}$$

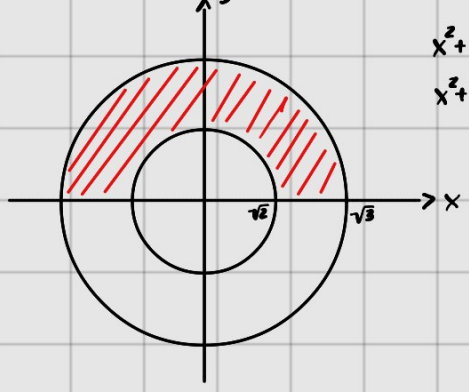
$u = 1+y^2$, $du = 2y\,dy$, $\begin{cases} y=0 \\ u=1 \end{cases}$, $\begin{cases} y=1 \\ u=2 \end{cases}$

$$\frac{1}{2}\int_1^2 \frac{\ln(u)}{u}\,du = \frac{1}{2}\left[\frac{\ln^2(u)}{2}\right]_1^2 = \frac{1}{2}\left[\frac{\ln^2(2)}{2} - \frac{\ln^2(1)}{2}\right] =$$

$$\int_0^1 dy \int_0^{y^2} \frac{y}{(1+x)(1+y^2)}\,dx = \frac{1}{2}\cdot\frac{\ln^2(2)}{2} = \frac{\ln^2(2)}{4}$$

ESERCIZIO

$\iint_D \sqrt{x^2+y^2}$ $\qquad D = \{2 \le x^2+y^2 \le 3 \; ; \; y \ge 0\}$



$x^2+y^2 \le 3 \longrightarrow x^2+y^2 = 3$

$x^2+y^2 \ge 2$

PARAMETRIZZAZIONE

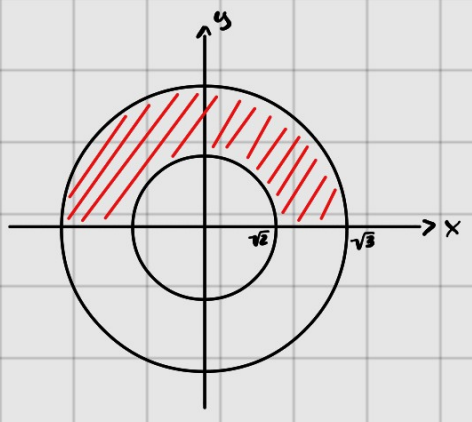
$$\begin{cases} x = \rho\cos\theta & \sqrt{2} < \rho < \sqrt{3} \\ y = \rho\operatorname{sen}\theta & 0 < \theta < \pi \end{cases}$$

$$\int_0^{\pi} d\theta \int_{\sqrt{2}}^{\sqrt{3}} \rho \cdot \sqrt{(\rho\cos\theta)^2 + (\rho\operatorname{sen}\theta)^2}\, d\rho = \int_0^{\pi} d\theta \int_{\sqrt{2}}^{\sqrt{3}} \rho \cdot \sqrt{\rho^2}\, d\rho = \int_0^{\pi} d\theta \int_{\sqrt{2}}^{\sqrt{3}} \rho^2 d\rho = \int_0^{\pi} d\theta \left[\frac{\rho^3}{3}\right]_{\sqrt{2}}^{\sqrt{3}} = \int_0^{\pi} d\theta \left(\frac{27}{3} - \frac{\sqrt{8}}{3}\right) = \frac{\sqrt{27}-\sqrt{8}}{3} \cdot \theta \Big|_0^{\pi} = \frac{\sqrt{27}-\sqrt{8}}{3}(\pi)$$

($\cos^2\theta + \operatorname{sen}^2\theta = 1$)

ESERCIZIO

$\iint_D (x^2 y)\, dx\, dy$ $\qquad D = \{1 \le x^2+y^2 \le 4 \; ; \; y \ge 0\}$



PARAMETRIZZAZIONE

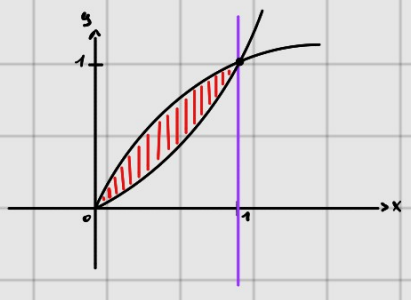
$$\begin{cases} x = \rho\cos\theta & 1 \le \rho \le 2 \\ y = \rho\operatorname{sen}\theta & 0 \le \theta \le \pi \end{cases}$$

$$\int_0^{\pi} d\theta \int_1^2 \rho^2\cos^2\theta \cdot \rho\operatorname{sen}\theta \cdot \rho\, d\rho = \int_0^{\pi} d\theta \int_1^2 \rho^4 \cos^2\theta \cdot \operatorname{sen}\theta\, d\rho =$$

$$= \int_0^{\pi} d\theta \int_1^2 \rho^4 (1-\operatorname{sen}^2\theta) \cdot \operatorname{sen}\theta\, d\rho$$

ESERCIZIO

$\iint_D (y\sqrt{x})\, dx\, dy$ $\qquad D = \{0 \le x \le 1 \; , \; x^2 \le y \le \sqrt{x}\}$



$$\int_0^1 \left(\int_{x^2}^{\sqrt{x}} y\sqrt{x}\, dy \right) dx = \int_0^1 \sqrt{x}\, dx \int_{x^2}^{\sqrt{x}} y\, dy = \int_0^1 \sqrt{x}\, dx \left[\frac{y^2}{2}\right]_{x^2}^{\sqrt{x}} =$$

$$\int_0^1 \sqrt{x}\, dx \left[\frac{(x^{\frac{1}{2}})^2}{2} - \frac{(x^2)^2}{2}\right] = \frac{x^4}{2} - \frac{x}{2} = \int_0^1 \sqrt{x}\left(\frac{x^4}{2} - \frac{x}{2}\right) dx = \int_0^1 x^{\frac{1}{2}}\left[\frac{x^4}{2} - \frac{x}{2}\right] dx = \int_0^1 \frac{x^{\frac{9}{2}}}{2} - \frac{x^{\frac{3}{2}}}{2}\, dx = \frac{1}{2}\int_0^1 x^{\frac{9}{2}} dx - \frac{1}{2}\int_0^1 x^{\frac{3}{2}} dx = \left[\frac{1}{2}\left(\frac{x^{\frac{11}{2}}}{\frac{11}{2}}\right) - \frac{1}{2}\left(\frac{x^{\frac{5}{2}}}{\frac{5}{2}}\right)\right]_0^1 =$$

$$= \left[\frac{1}{2}\left(x^{\frac{11}{2}} \cdot \frac{2}{11} - \frac{1}{2} \cdot x^{\frac{5}{2}} \cdot \frac{2}{5}\right)\right]_0^1 = \left[\frac{1}{2} \cdot \frac{2}{11} - \frac{1}{2} \cdot \frac{2}{5}\right] - 0 = \frac{1}{11} - \frac{1}{5} = \frac{5-11}{55} = -\frac{6}{55}$$